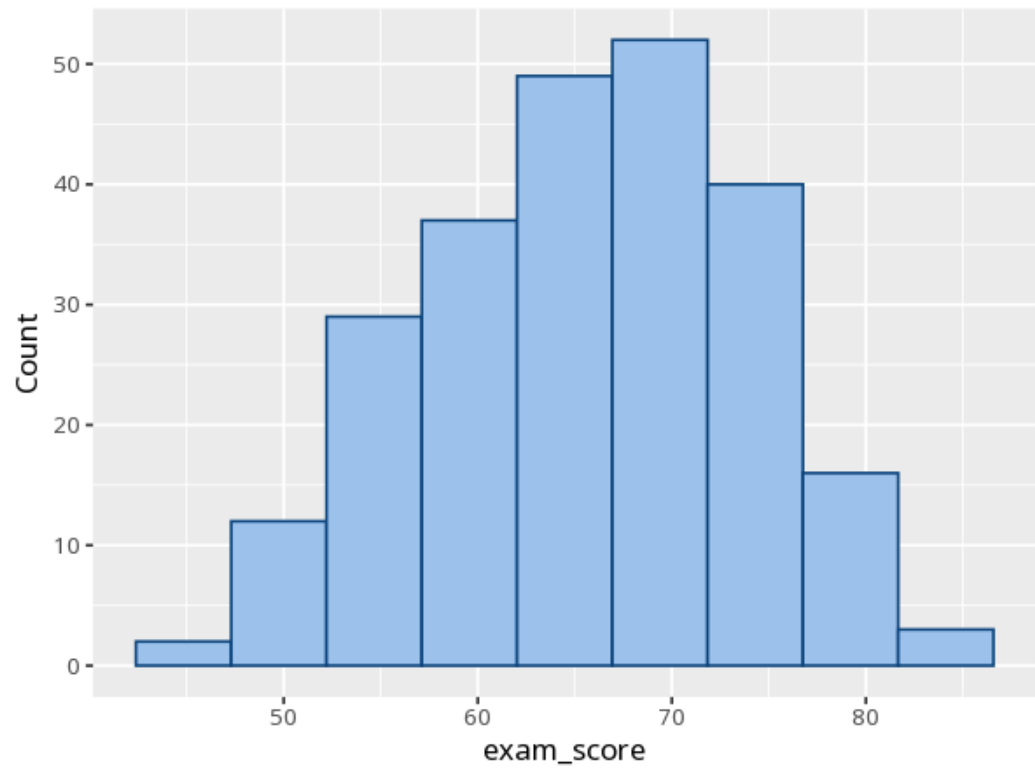
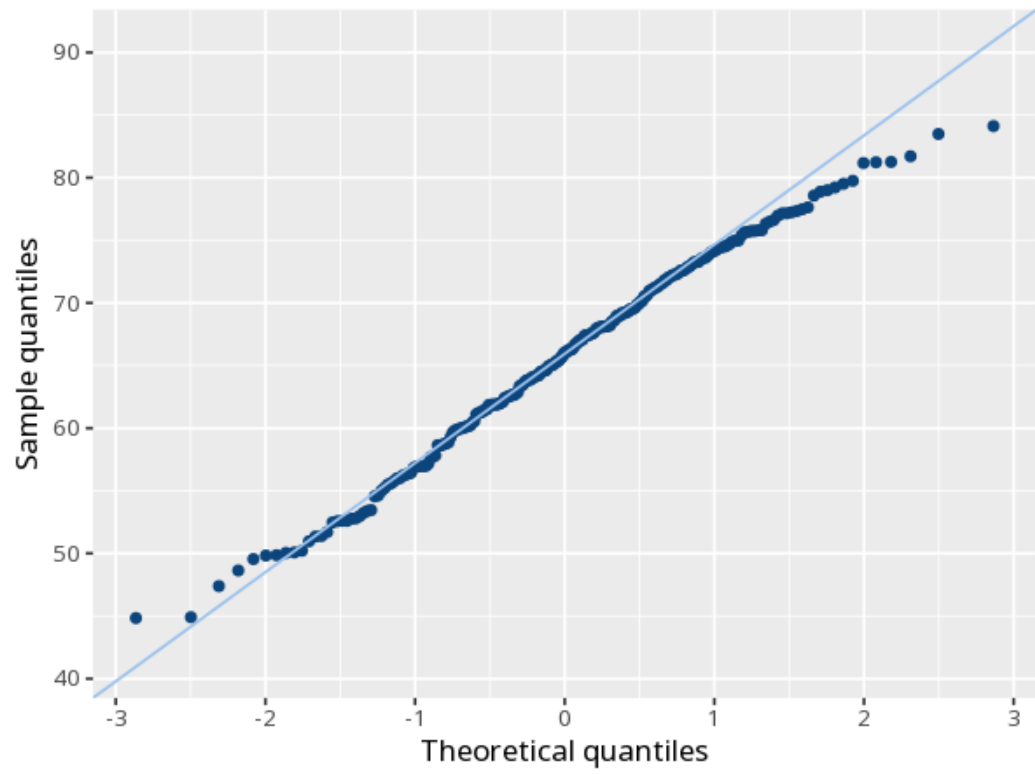
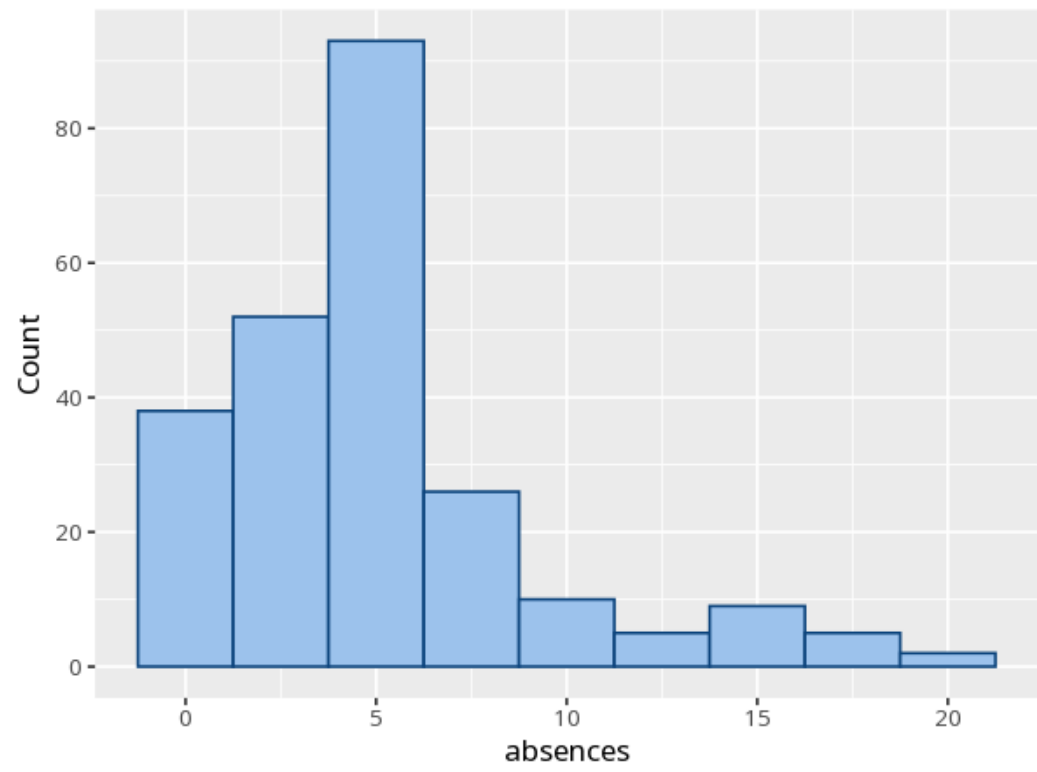


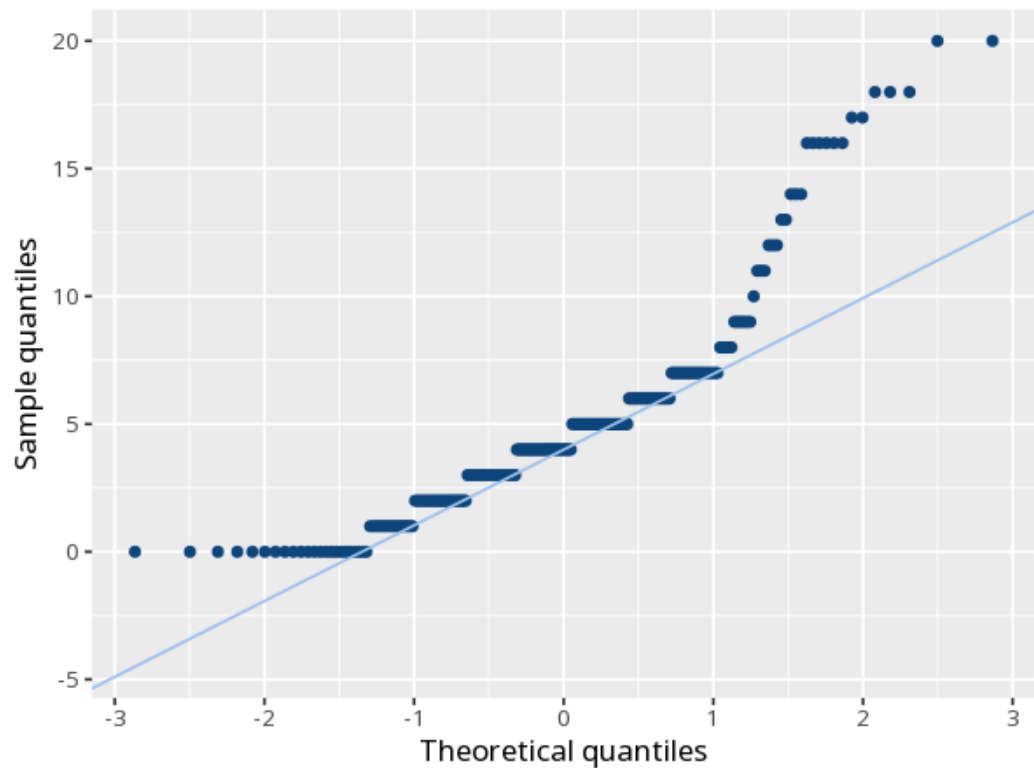
1 Distribution & normality

Variable	Test	Statistic	N	p	Skew	Kurtosis (excess)
exam_score	Shapiro-Wilk	W 0.99	240	.158	−0.18	−0.53
exam_score	K-S (Lilliefors)	D 0.04	240	.608	−0.18	−0.53
absences	Shapiro-Wilk	W 0.86	240	<.001	1.50	2.43
absences	K-S (Lilliefors)	D 0.17	240	<.001	1.50	2.43









A small p (below .05) suggests the data depart from normality. With large samples these tests over-flag trivial deviations; with small samples they have low power, so a $p > .05$ does not prove normality — it only means no departure was detected. Either way, judge normality mainly from the Q-Q plot — points hugging the diagonal indicate normality.

exam_score: Normality was assessed with the Shapiro-Wilk test, $W = .99$, $p = .158$. absences: Normality was assessed with the Shapiro-Wilk test, $W = .86$, $p < .001$.

Statistical basis: Shapiro-Wilk test for normality (3-5000 cases) (Shapiro, S. S., Wilk, M. B. (1965). "An analysis of variance test for normality (complete samples)." *Biometrika*, 52(3/4), 591-611.); Additional normality tests (Lilliefors and related) (Gross J, Ligges U (2015). *nortest: Tests for Normality*. R package.).